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Here’s how I approach messy, ambiguous, real-world data and extract strategic insight

FROM RAW DATA TO STRATEGY: UNDERSTANDING WHAT DRIVES MUSIC PERFORMANCE

MULTI-SOURCE ANALYSIS COMBINING CHARTS, LYRICS, SENTIMENT, AND AUDIO FEATURES

2026

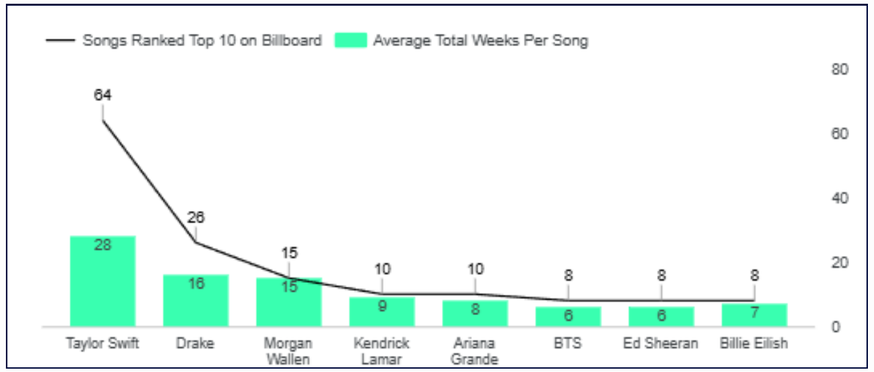
WHAT MAKES A SONG CHART? EVIDENCE FROM TAYLOR SWIFT'S CATALOG

Why Taylor Swift's catalog?

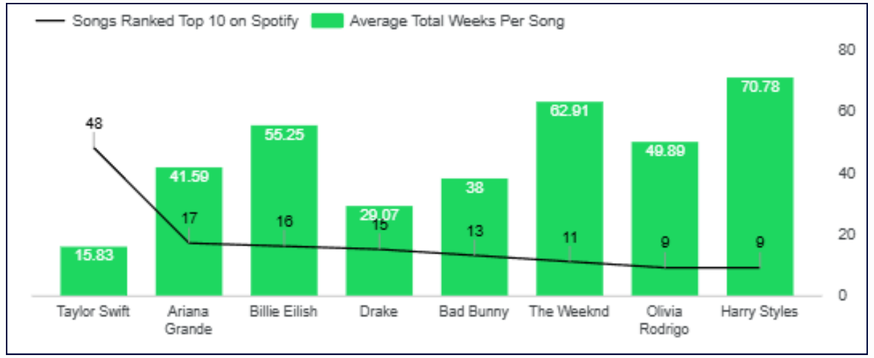
- 20 years of data across multiple industry and genre shifts
- Natural industry experiments (Spotify boycott 2014-2017, re-releases)
- Cross-platform visibility (US vs Global)

This tests whether her huge **success is driven by the music itself or by external factors**

Context: Taylor Swift vs industry peers



Artists with Most Top 10 Hits (Billboard Hot 100)



Artists with Most Top 10 Hits (Spotify Global 200)

SUMMARY: AUDIO FEATURES SHOW LIMITED PREDICTIVE POWER; PERFORMANCE IS DRIVEN BY STRATEGY

- **Audio features do not differentiate hits.** Tempo, energy, and danceability show no consistent relationship with peak performance. Top songs exist across the full range of these characteristics.

- **Emotional tone influences how songs perform, not just if they perform.** Negative songs peak higher, while neutral songs sustain longer on streaming platforms, indicating different emotional profiles drive peak vs longevity.

- **Chart performance is shaped by system-level dynamics.** Release timing, cultural moments, and fan mobilization explain more variation than any musical feature. Success is a product of context, not composition.

METHODOLOGY: 4 DATASETS, 1 BUILT FROM SCRATCH

Data Used in BigQuery

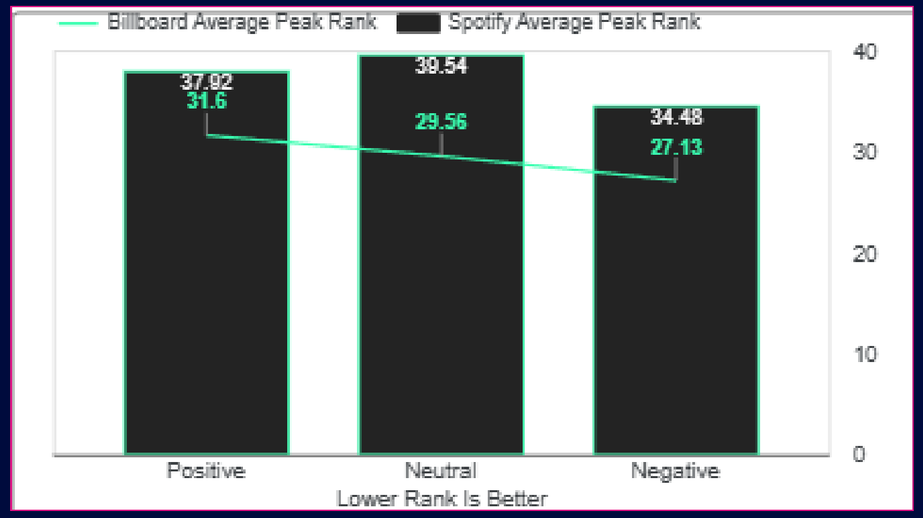
Dataset	Source
Billboard Hot 100 (US) 1958-Oct 2025	Scraped via Python with AI-assisted development
Spotify Global 200 2017-Oct 2025	Kaggle + manual extensions (Spotify scraping limits)
Taylor Swift's Lyrical Characteristics	Custom dataset: manual lyrics (genius.com), engineered lyrical characteristics, sentiment
Spotify Audio Features	Spotify API (pre-2024 access)

Sentiment Score Calculation Process

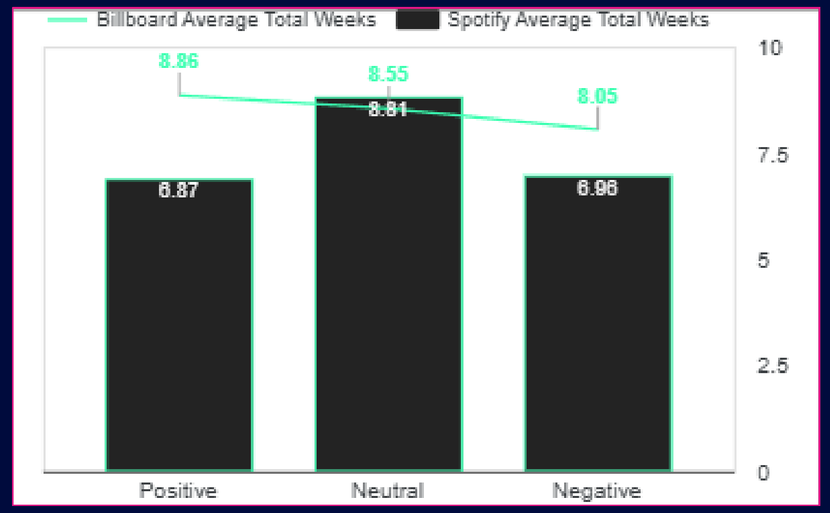
- Lyrics sourced manually and processed in Python. Sentiment classified using CardiffNLP RoBERTa (context-aware model)
- ~18% of songs were adjusted via human-in-the-loop validation to correct model misclassification (*e.g. misunderstood metaphors, context*).

DIFFERENT EMOTIONAL PROFILES DRIVE PEAK PERFORMANCE VS LONGEVITY

- Negative songs achieve stronger peak positions than positive songs across both platforms (*Billboard: 27.13 vs 31.6; Spotify: 34.48 vs 37.92 — lower rank is better)*
- Longevity diverges by platform: neutral sentiment lasts longest on Spotify, while positive performs best on Billboard
- Only 1 of 13 Billboard (and 1 of 7 Spotify) #1 hits carry positive sentiment



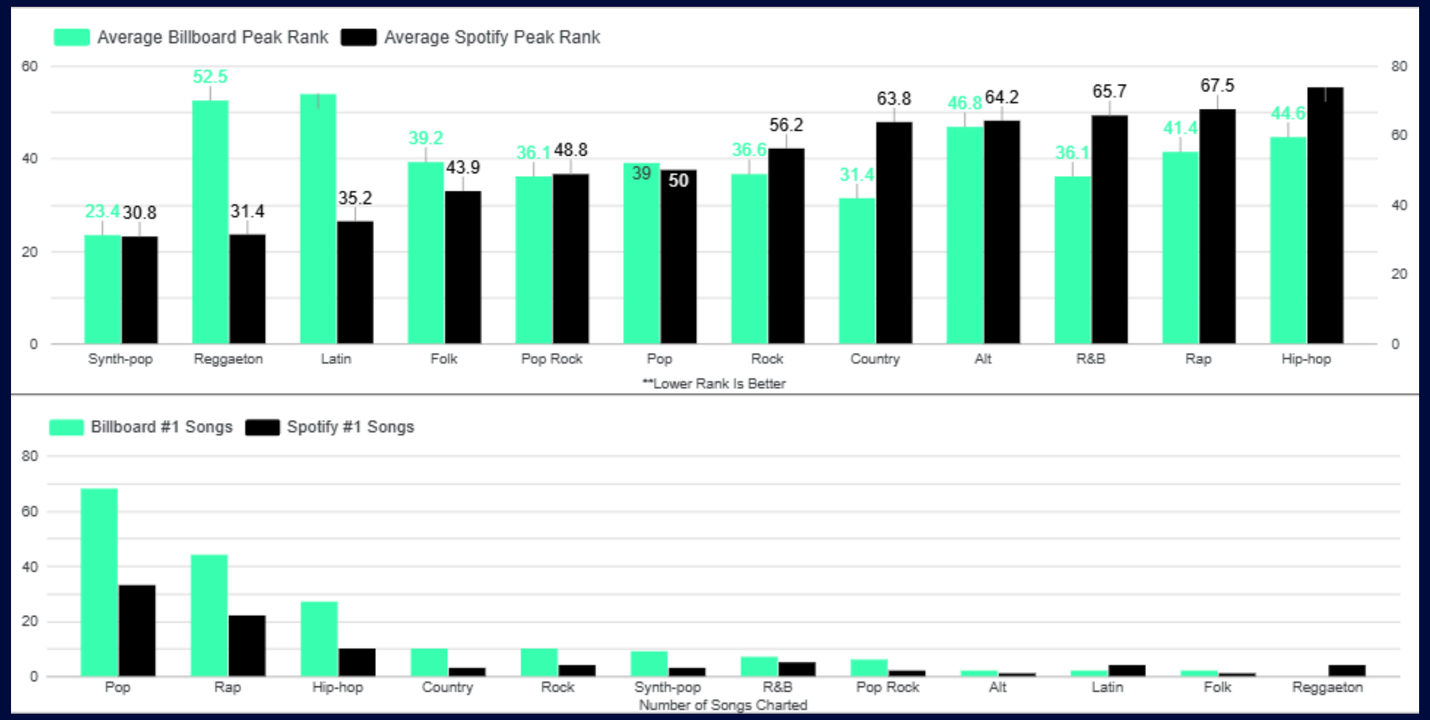
Billboard Hot 100 & Spotify Global Average Peak Ranks by Sentiment



Billboard Hot 100 & Spotify Global Average Total Weeks by Sentiment

BILLBOARD HOT 100 AND SPOTIFY GLOBAL REFLECT DIFFERENT AUDIENCES AND CONSUMPTION PATTERNS

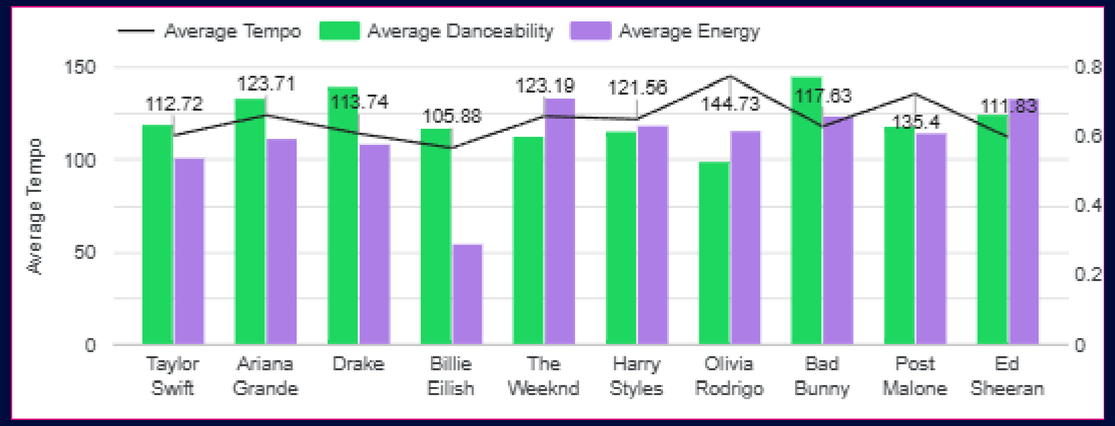
- Country performs strongly on Billboard but weakly on Spotify *(31.4 vs 63.8 — US-centric vs global gap)*
- Reggaeton shows the opposite pattern: weak on Billboard but strong on Spotify
- Genre performance varies significantly by platform, reflecting differences between US radio-driven and global streaming audiences



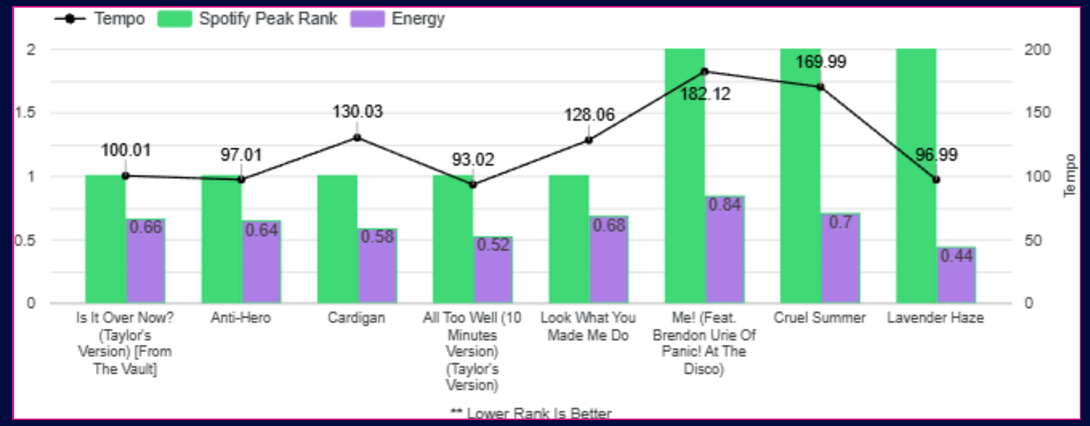
Genre Performance on Both Platforms (By Average Rank and Number of #1 Hits)

MOST MUSICAL FEATURES ARE BASELINE CONDITIONS, NOT DIFFERENTIATORS

- Top and lower-performing songs are not distinguishable by audio or structural features
- Features like Tempo, Energy, and Song Structure are necessary but not necessarily sufficient for success
- Performance differences emerge from context, not composition



Average Audio Features of Top-charting Spotify Artists

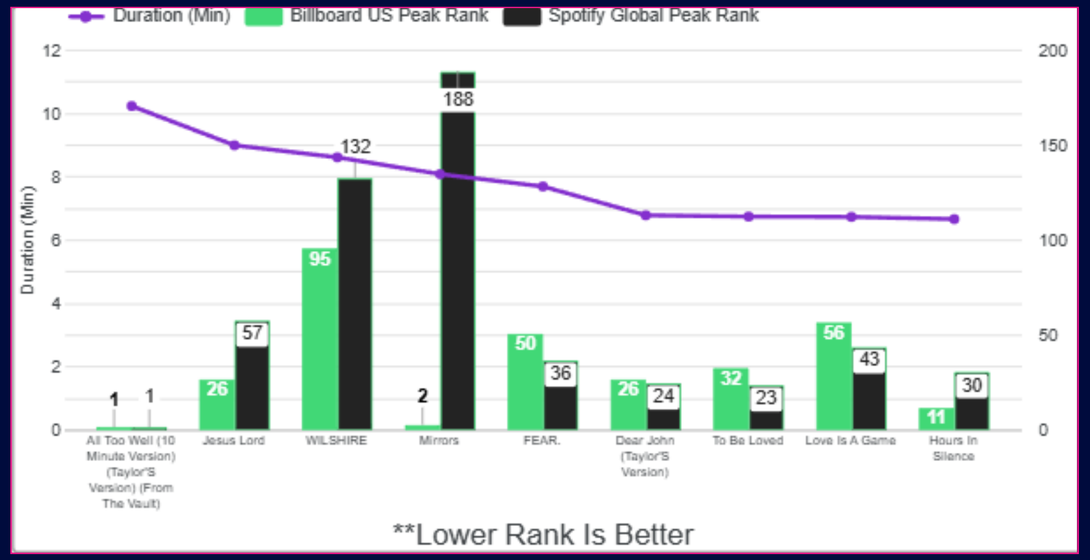


Audio Features of Swift's Top Tracks by Peak Rank

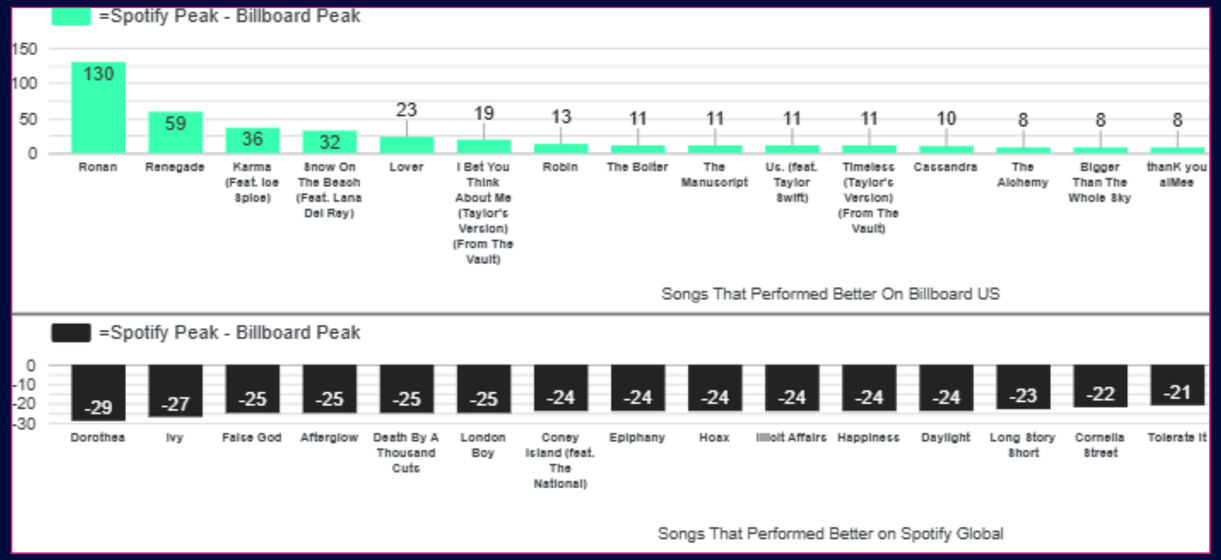
THE BIGGEST PERFORMANCE OUTLIERS ARE DRIVEN BY CULTURAL MOMENTS MORE THAN SONG FEATURES

- All Too Well (10 Minute Version) reached #1 because its 10-minute structure became a cultural event, amplified by fan mobilization

- The same song can succeed differently across platforms, reflecting US radio-driven vs global streaming dynamic. *(e.g., Dorothea & Ivy (Evermore deep cuts) saw significant Spotify uplift, driven by TikTok fan edits)*



Songs' Chart Performance Ordered By Duration (min)



Billboard Hot 100 & Spotify Global Taylor Songs' Peak Rank Difference

LIMITATIONS & NEXT STEPS

- **Data coverage gap:** Spotify data begins in 2017 due to catalog removal, limiting comparability with pre-streaming era performance; incorporating Apple Music or historical radio data would improve coverage. Audio Features are unavailable for public use past-2024.

- **Model limitations**: Sentiment classification relied on RoBERTa with ~18% manual adjustment; more advanced, context-aware models or multi-class sentiment frameworks would improve accuracy.

- **Unobserved variables** (such as marketing spend, release strategy, playlist placement, and tour cycles) are not captured, but likely explain more variance than measured features.

- **Next step**: Integrate cross-platform and marketing data, move from descriptive analysis to predictive modeling of chart performance.

IF I HAD LIMITLESS RESOURCES & WHAT I ACTUALLY LEARNED

What I'd build with more resources (including time)

- Integrate external drivers: TikTok trends, Eras Tour timing, and playlist placement to capture cultural momentum (*e.g. Cruel Summer's delayed #1 is largely invisible without this data)*

- Test whether findings generalize across artists vs being Taylor-specific

What this project actually taught me

- Understanding context *(e.g. "happiness" as a sad song)* directly impacts analytics
- Not all data is useful; several features *(e.g. temporal words, lyrical density)* were removed due to low signal-to-noise

The key takeaway: The biggest drivers weren't in the dataset. Strategy, timing, tours might explain more than measurable song features

HITS ARE DRIVEN BY CONTEXT, NOT JUST THE MUSIC

Audio features, tempo, and danceability showed no consistent relationship with peak rank. The variables that explained the most variance (release timing, playlist placement, cultural momentum) weren't in the dataset at all.

EXPLORE THE FULL ANALYSIS:

Looker Studio Dashboard GitHub

CONNECT WITH ME:

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